

Static Analysis and Verification

SAV 2021/2022

Tomáš Vojnar

`vojnar@fit.vutbr.cz`

**Brno University of Technology
Faculty of Information Technology
Božetěchova 2, 612 66 Brno**

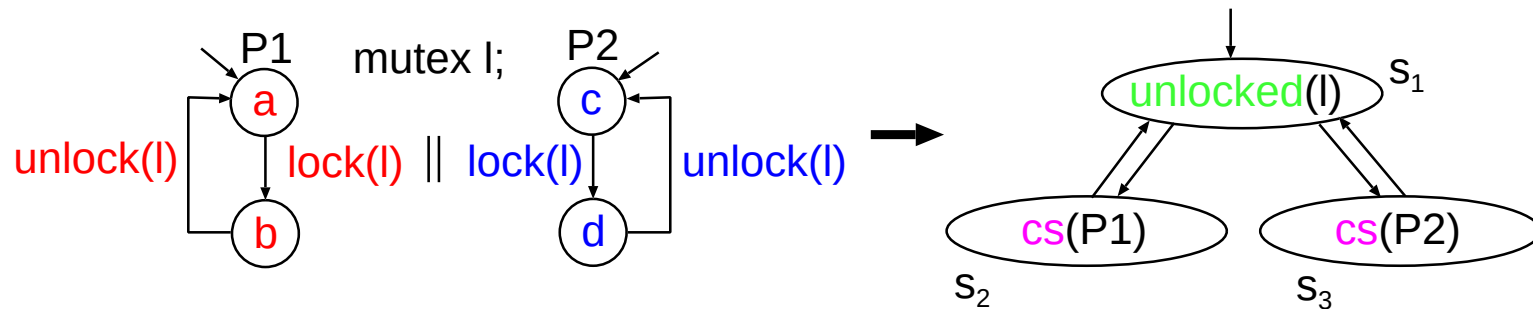
Temporal Logics:

CTL*, CTL, LTL

Model of Computation

Kripke Structures

- ❖ Informally, **Kripke structures** are oriented graphs whose
 - **vertices** correspond to configurations of the examined system,
 - the vertices are **labelled** by atomic propositions that are true in the appropriate configurations, and
 - **edges** encode possible transitions between the configurations.



- ❖ Can be **generated** (or at least defined and used implicitly) from the source description of examined systems.
- ❖ The generation involves the **state explosion problem**, or the Kripke structure may be **infinite**—in the following, we, however, concentrate on finite Kripke structures.

Kripke Structures

- ❖ Let AP be a set of **atomic propositions** about the configurations of the examined system.
- ❖ Formally, a (finite) **Kripke structure** M over AP is a tuple $M = (S, S_0, R, L)$ where
 - S is a finite set of **states**,
 - $S_0 \subseteq S$ is a set of **initial states**,
 - $R \subseteq S \times S$ is a **transition relation**, for convenience supposed to be total (i.e. $\forall s \in S \exists s' \in S. R(s, s')$),
 - $L : S \rightarrow 2^{AP}$ is a **labelling function** that labels each state by the set of atomic propositions that are true in it.

Kripke Structures

- ❖ Let AP be a set of **atomic propositions** about the configurations of the examined system.
- ❖ Formally, a (finite) **Kripke structure** M over AP is a tuple $M = (S, S_0, R, L)$ where
 - S is a finite set of **states**,
 - $S_0 \subseteq S$ is a set of **initial states**,
 - $R \subseteq S \times S$ is a **transition relation**, for convenience supposed to be total (i.e. $\forall s \in S \exists s' \in S. R(s, s')$),
 - $L : S \rightarrow 2^{AP}$ is a **labelling function** that labels each state by the set of atomic propositions that are true in it.
- ❖ For the example from the previous slide, we have:
 - $AP = \{unlocked(l), cs(P1), cs(P2)\}$,
 - $S = \{s_1, s_2, s_3\}$,
 - $S_0 = \{s_1\}$,
 - $R = \{(s_1, s_2), (s_2, s_1), (s_1, s_3), (s_3, s_1)\}$,
 - $L = \{(s_1, \{unlocked(l)\}), (s_2, \{cs(P1)\}), (s_3, \{cs(P2)\})\}$.

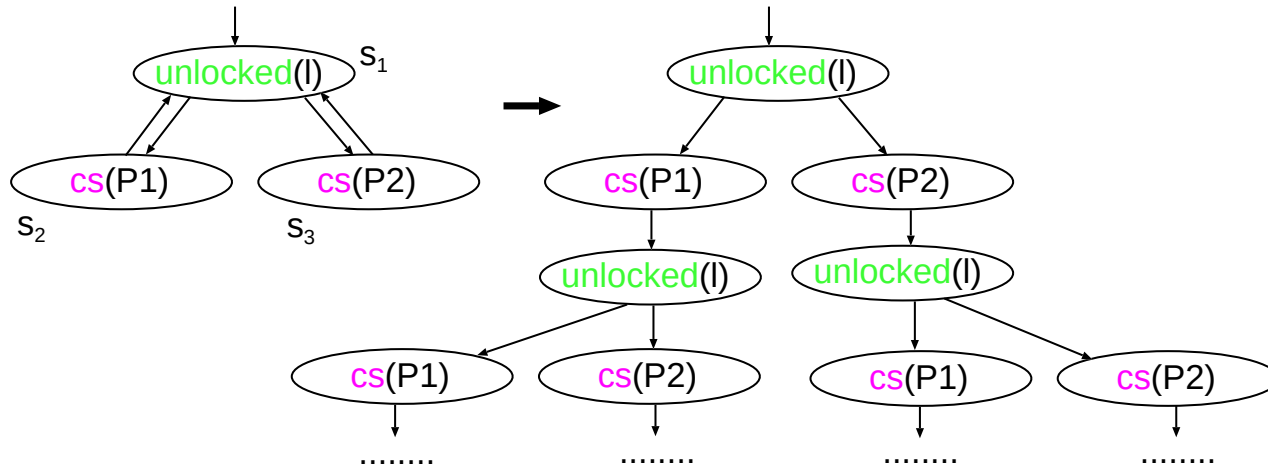
Kripke Structures

- ❖ A **path** π in a Kripke structure M is an infinite sequence of states $\pi = s_0 s_1 s_2 \dots$ such that $\forall i \in \mathbb{N}. R(s_i, s_{i+1})$.
- ❖ We denote $\Pi(M, s)$ the set of all paths in M that start at $s \in S$.
- ❖ The **suffix** π^i of a path $\pi = s_0 s_1 s_2 \dots s_i s_{i+1} s_{i+2} \dots$ is the path $\pi^i = s_i s_{i+1} s_{i+2} \dots$ starting at s_i .

The CTL* Logic

CTL*—Basic Idea

- ❖ CTL* formulae describe properties of **computation trees**.
- ❖ Infinite computation trees are obtained by **unwinding** a Kripke structure from its initial states.



- ❖ CTL* formulae **consist of**:
 - atomic propositions,
 - Boolean connectives,
 - path quantifiers,
 - temporal operators.

CTL*—Quantifiers and Operators

❖ **Path quantifiers**—describe the branching structure of a computation tree:

- E : for some computation path leading from a state,
- A : for all computation paths leading from a state.

❖ **Temporal operators**—properties of a path through a computation tree:

- $X \varphi$ (“**next time**”, $\bigcirc$): the property φ holds in the second state of the path,
- $F \varphi$ (“**eventually**” or “**sometimes**”, $\Diamond$): the property φ holds at some state of the path,
- $G \varphi$ (“**always**” or “**globally**”, $\Box$): the property φ holds at all states of the path,
- $\varphi U \psi$ (“**until**”): the property ψ holds at some state of the path, and the property φ holds at all preceding states of the path,
- $\varphi R \psi$ (“**release**”): the property ψ holds at all states of the path up to (and including) the first state where the property φ holds (if such a state exists).

CTL*—The Syntax

- ❖ Let AP be a non-empty set of atomic propositions.
- ❖ The syntax of **state formulae**, which are true in a specific state, is given by the following rules:
 - If $p \in AP$, then p is a state formula.
 - If φ and ψ are state formulae, then $\neg\varphi$, $\varphi \vee \psi$, $\varphi \wedge \psi$ are state formulae.
 - If φ is a path formula, then $E\varphi$ and $A\varphi$ are state formulae.
- ❖ The syntax of **path formulae**, which are true along a specific path, is given by the following rules:
 - If φ is a state formula, then φ is a path formula too.
 - If φ and ψ are path formulae, then $\neg\varphi$, $\varphi \vee \psi$, $\varphi \wedge \psi$, $X\varphi$, $F\varphi$, $G\varphi$, $\varphi U\psi$, and $\varphi R\psi$ are path formulae.
- ❖ **CTL*** is the set of **state formulae** generated by the above rules.

CTL*—The Semantics

- ❖ Let a Kripke structure $M = (S, S_0, R, L)$ over a set of atomic propositions AP be given.
- ❖ For a *state formula* φ over AP , we denote $M, s \models \varphi$ the fact that φ holds at $s \in S$.
- ❖ For a *path formula* φ over AP , we denote $M, \pi \models \varphi$ the fact that φ holds along a path π in M .

CTL*—The Semantics

- ❖ Let a Kripke structure $M = (S, S_0, R, L)$ over a set of atomic propositions AP be given.
- ❖ For a *state formula* φ over AP , we denote $M, s \models \varphi$ the fact that φ holds at $s \in S$.
- ❖ For a *path formula* φ over AP , we denote $M, \pi \models \varphi$ the fact that φ holds along a path π in M .
- ❖ Let $s \in S$, π be a path in M , φ_1, φ_2 be state formulae over AP , $p \in AP$, and ψ_1, ψ_2 be path formulae over AP . We **define the relation $\models$ inductively** as follows:

Continued at the next slide...

CTL*—The Semantics

- ❖ Let a Kripke structure $M = (S, S_0, R, L)$ over a set of atomic propositions AP be given.
- ❖ For a *state formula* φ over AP , we denote $M, s \models \varphi$ the fact that φ holds at $s \in S$.
- ❖ For a *path formula* φ over AP , we denote $M, \pi \models \varphi$ the fact that φ holds along a path π in M .
- ❖ Let $s \in S$, π be a path in M , φ_1, φ_2 be state formulae over AP , $p \in AP$, and ψ_1, ψ_2 be path formulae over AP . We **define the relation $\models$ inductively** as follows:
 - $M, s \models p$ iff $p \in L(s)$.

Continued at the next slide...

CTL*—The Semantics

- ❖ Let a Kripke structure $M = (S, S_0, R, L)$ over a set of atomic propositions AP be given.
- ❖ For a *state formula* φ over AP , we denote $M, s \models \varphi$ the fact that φ holds at $s \in S$.
- ❖ For a *path formula* φ over AP , we denote $M, \pi \models \varphi$ the fact that φ holds along a path π in M .
- ❖ Let $s \in S$, π be a path in M , φ_1, φ_2 be state formulae over AP , $p \in AP$, and ψ_1, ψ_2 be path formulae over AP . We **define the relation $\models$ inductively** as follows:
 - $M, s \models p$ iff $p \in L(s)$.
 - $M, s \models \neg\varphi_1$ iff $M, s \not\models \varphi_1$.

Continued at the next slide...

CTL*—The Semantics

- ❖ Let a Kripke structure $M = (S, S_0, R, L)$ over a set of atomic propositions AP be given.
- ❖ For a *state formula* φ over AP , we denote $M, s \models \varphi$ the fact that φ holds at $s \in S$.
- ❖ For a *path formula* φ over AP , we denote $M, \pi \models \varphi$ the fact that φ holds along a path π in M .
- ❖ Let $s \in S$, π be a path in M , φ_1, φ_2 be state formulae over AP , $p \in AP$, and ψ_1, ψ_2 be path formulae over AP . We **define the relation $\models$ inductively** as follows:
 - $M, s \models p$ iff $p \in L(s)$.
 - $M, s \models \neg\varphi_1$ iff $M, s \not\models \varphi_1$.
 - $M, s \models \varphi_1 \vee \varphi_2$ iff $M, s \models \varphi_1$ or $M, s \models \varphi_2$.

Continued at the next slide...

CTL*—The Semantics

- ❖ Let a Kripke structure $M = (S, S_0, R, L)$ over a set of atomic propositions AP be given.
- ❖ For a *state formula* φ over AP , we denote $M, s \models \varphi$ the fact that φ holds at $s \in S$.
- ❖ For a *path formula* φ over AP , we denote $M, \pi \models \varphi$ the fact that φ holds along a path π in M .
- ❖ Let $s \in S$, π be a path in M , φ_1, φ_2 be state formulae over AP , $p \in AP$, and ψ_1, ψ_2 be path formulae over AP . We **define the relation $\models$ inductively** as follows:
 - $M, s \models p$ iff $p \in L(s)$.
 - $M, s \models \neg\varphi_1$ iff $M, s \not\models \varphi_1$.
 - $M, s \models \varphi_1 \vee \varphi_2$ iff $M, s \models \varphi_1$ or $M, s \models \varphi_2$.
 - $M, s \models \varphi_1 \wedge \varphi_2$ iff $M, s \models \varphi_1$ and $M, s \models \varphi_2$.

Continued at the next slide...

CTL*—The Semantics

- ❖ Let a Kripke structure $M = (S, S_0, R, L)$ over a set of atomic propositions AP be given.
- ❖ For a *state formula* φ over AP , we denote $M, s \models \varphi$ the fact that φ holds at $s \in S$.
- ❖ For a *path formula* φ over AP , we denote $M, \pi \models \varphi$ the fact that φ holds along a path π in M .
- ❖ Let $s \in S$, π be a path in M , φ_1, φ_2 be state formulae over AP , $p \in AP$, and ψ_1, ψ_2 be path formulae over AP . We **define the relation $\models$ inductively** as follows:
 - $M, s \models p$ iff $p \in L(s)$.
 - $M, s \models \neg\varphi_1$ iff $M, s \not\models \varphi_1$.
 - $M, s \models \varphi_1 \vee \varphi_2$ iff $M, s \models \varphi_1$ or $M, s \models \varphi_2$.
 - $M, s \models \varphi_1 \wedge \varphi_2$ iff $M, s \models \varphi_1$ and $M, s \models \varphi_2$.
 - $M, s \models E \psi_1$ iff $\exists \pi \in \Pi(M, s). M, \pi \models \psi_1$.

Continued at the next slide...

CTL*—The Semantics

- ❖ Let a Kripke structure $M = (S, S_0, R, L)$ over a set of atomic propositions AP be given.
- ❖ For a *state formula* φ over AP , we denote $M, s \models \varphi$ the fact that φ holds at $s \in S$.
- ❖ For a *path formula* φ over AP , we denote $M, \pi \models \varphi$ the fact that φ holds along a path π in M .
- ❖ Let $s \in S$, π be a path in M , φ_1, φ_2 be state formulae over AP , $p \in AP$, and ψ_1, ψ_2 be path formulae over AP . We **define the relation $\models$ inductively** as follows:
 - $M, s \models p$ iff $p \in L(s)$.
 - $M, s \models \neg\varphi_1$ iff $M, s \not\models \varphi_1$.
 - $M, s \models \varphi_1 \vee \varphi_2$ iff $M, s \models \varphi_1$ or $M, s \models \varphi_2$.
 - $M, s \models \varphi_1 \wedge \varphi_2$ iff $M, s \models \varphi_1$ and $M, s \models \varphi_2$.
 - $M, s \models E \psi_1$ iff $\exists \pi \in \Pi(M, s). M, \pi \models \psi_1$.
 - $M, s \models A \psi_1$ iff $\forall \pi \in \Pi(M, s). M, \pi \models \psi_1$.

Continued at the next slide...

CTL*—The Semantics

Continued from the previous slide...

- $M, \pi \models \varphi_1$ iff $M, s_0 \models \varphi_1$ where s_0 is the first state of π .

CTL*—The Semantics

Continued from the previous slide...

- $M, \pi \models \varphi_1$ iff $M, s_0 \models \varphi_1$ where s_0 is the first state of π .
- $M, \pi \models \neg\psi_1$ iff $M, \pi \not\models \psi_1$.

CTL*—The Semantics

Continued from the previous slide...

- $M, \pi \models \varphi_1$ iff $M, s_0 \models \varphi_1$ where s_0 is the first state of π .
- $M, \pi \models \neg\psi_1$ iff $M, \pi \not\models \psi_1$.
- $M, \pi \models \psi_1 \vee \psi_2$ iff $M, \pi \models \psi_1$ or $M, \pi \models \psi_2$.

CTL*—The Semantics

Continued from the previous slide...

- $M, \pi \models \varphi_1$ iff $M, s_0 \models \varphi_1$ where s_0 is the first state of π .
- $M, \pi \models \neg\psi_1$ iff $M, \pi \not\models \psi_1$.
- $M, \pi \models \psi_1 \vee \psi_2$ iff $M, \pi \models \psi_1$ or $M, \pi \models \psi_2$.
- $M, \pi \models \psi_1 \wedge \psi_2$ iff $M, \pi \models \psi_1$ and $M, \pi \models \psi_2$.

CTL*—The Semantics

Continued from the previous slide...

- $M, \pi \models \varphi_1$ iff $M, s_0 \models \varphi_1$ where s_0 is the first state of π .
- $M, \pi \models \neg\psi_1$ iff $M, \pi \not\models \psi_1$.
- $M, \pi \models \psi_1 \vee \psi_2$ iff $M, \pi \models \psi_1$ or $M, \pi \models \psi_2$.
- $M, \pi \models \psi_1 \wedge \psi_2$ iff $M, \pi \models \psi_1$ and $M, \pi \models \psi_2$.
- $M, \pi \models X \psi_1$ iff $M, \pi^1 \models \psi_1$.

CTL*—The Semantics

Continued from the previous slide...

- $M, \pi \models \varphi_1$ iff $M, s_0 \models \varphi_1$ where s_0 is the first state of π .
- $M, \pi \models \neg\psi_1$ iff $M, \pi \not\models \psi_1$.
- $M, \pi \models \psi_1 \vee \psi_2$ iff $M, \pi \models \psi_1$ or $M, \pi \models \psi_2$.
- $M, \pi \models \psi_1 \wedge \psi_2$ iff $M, \pi \models \psi_1$ and $M, \pi \models \psi_2$.
- $M, \pi \models X \psi_1$ iff $M, \pi^1 \models \psi_1$.
- $M, \pi \models F \psi_1$ iff $\exists i \geq 0. M, \pi^i \models \psi_1$.

CTL*—The Semantics

Continued from the previous slide...

- $M, \pi \models \varphi_1$ iff $M, s_0 \models \varphi_1$ where s_0 is the first state of π .
- $M, \pi \models \neg\psi_1$ iff $M, \pi \not\models \psi_1$.
- $M, \pi \models \psi_1 \vee \psi_2$ iff $M, \pi \models \psi_1$ or $M, \pi \models \psi_2$.
- $M, \pi \models \psi_1 \wedge \psi_2$ iff $M, \pi \models \psi_1$ and $M, \pi \models \psi_2$.
- $M, \pi \models X \psi_1$ iff $M, \pi^1 \models \psi_1$.
- $M, \pi \models F \psi_1$ iff $\exists i \geq 0. M, \pi^i \models \psi_1$.
- $M, \pi \models G \psi_1$ iff $\forall i \geq 0. M, \pi^i \models \psi_1$.

CTL*—The Semantics

Continued from the previous slide...

- $M, \pi \models \varphi_1$ iff $M, s_0 \models \varphi_1$ where s_0 is the first state of π .
- $M, \pi \models \neg\psi_1$ iff $M, \pi \not\models \psi_1$.
- $M, \pi \models \psi_1 \vee \psi_2$ iff $M, \pi \models \psi_1$ or $M, \pi \models \psi_2$.
- $M, \pi \models \psi_1 \wedge \psi_2$ iff $M, \pi \models \psi_1$ and $M, \pi \models \psi_2$.
- $M, \pi \models X \psi_1$ iff $M, \pi^1 \models \psi_1$.
- $M, \pi \models F \psi_1$ iff $\exists i \geq 0. M, \pi^i \models \psi_1$.
- $M, \pi \models G \psi_1$ iff $\forall i \geq 0. M, \pi^i \models \psi_1$.
- $M, \pi \models \psi_1 U \psi_2$ iff $\exists i \geq 0. M, \pi^i \models \psi_2$ and $\forall 0 \leq j < i. M, \pi^j \models \psi_1$.

CTL*—The Semantics

Continued from the previous slide...

- $M, \pi \models \varphi_1$ iff $M, s_0 \models \varphi_1$ where s_0 is the first state of π .
- $M, \pi \models \neg\psi_1$ iff $M, \pi \not\models \psi_1$.
- $M, \pi \models \psi_1 \vee \psi_2$ iff $M, \pi \models \psi_1$ or $M, \pi \models \psi_2$.
- $M, \pi \models \psi_1 \wedge \psi_2$ iff $M, \pi \models \psi_1$ and $M, \pi \models \psi_2$.
- $M, \pi \models X \psi_1$ iff $M, \pi^1 \models \psi_1$.
- $M, \pi \models F \psi_1$ iff $\exists i \geq 0. M, \pi^i \models \psi_1$.
- $M, \pi \models G \psi_1$ iff $\forall i \geq 0. M, \pi^i \models \psi_1$.
- $M, \pi \models \psi_1 U \psi_2$ iff $\exists i \geq 0. M, \pi^i \models \psi_2$ and $\forall 0 \leq j < i. M, \pi^j \models \psi_1$.
- $M, \pi \models \psi_1 R \psi_2$ iff $\forall i \geq 0. (\forall 0 \leq j < i. M, \pi^j \not\models \psi_1) \Rightarrow M, \pi^i \models \psi_2$.

CTL*—The Semantics

Continued from the previous slide...

- $M, \pi \models \varphi_1$ iff $M, s_0 \models \varphi_1$ where s_0 is the first state of π .
- $M, \pi \models \neg\psi_1$ iff $M, \pi \not\models \psi_1$.
- $M, \pi \models \psi_1 \vee \psi_2$ iff $M, \pi \models \psi_1$ or $M, \pi \models \psi_2$.
- $M, \pi \models \psi_1 \wedge \psi_2$ iff $M, \pi \models \psi_1$ and $M, \pi \models \psi_2$.
- $M, \pi \models X \psi_1$ iff $M, \pi^1 \models \psi_1$.
- $M, \pi \models F \psi_1$ iff $\exists i \geq 0. M, \pi^i \models \psi_1$.
- $M, \pi \models G \psi_1$ iff $\forall i \geq 0. M, \pi^i \models \psi_1$.
- $M, \pi \models \psi_1 U \psi_2$ iff $\exists i \geq 0. M, \pi^i \models \psi_2$ and $\forall 0 \leq j < i. M, \pi^j \models \psi_1$.
- $M, \pi \models \psi_1 R \psi_2$ iff $\forall i \geq 0. (\forall 0 \leq j < i. M, \pi^j \not\models \psi_1) \Rightarrow M, \pi^i \models \psi_2$.

❖ For a (state) CTL* formula φ , we write $M \models \varphi$ iff $\forall s_0 \in S_0. M, s_0 \models \varphi$.

CTL*—Basic Operators

❖ Provided that $AP \neq \emptyset$, it is easy to see that all CTL* operators can be derived from $\vee, \neg, X, U$, and E :

- let $p \in AP$, $true \equiv$ (and $false \equiv \neg true$),
- $\varphi \wedge \psi \equiv$
- $F \varphi \equiv$
- $G \varphi \equiv$
- $\varphi R \psi \equiv$
- $A \varphi \equiv$

CTL*—Basic Operators

❖ Provided that $AP \neq \emptyset$, it is easy to see that all CTL* operators can be derived from $\vee, \neg, X, U$, and E :

- let $p \in AP$, $true \equiv p \vee \neg p$ (and $false \equiv \neg true$),
- $\varphi \wedge \psi \equiv$
- $F \varphi \equiv$
- $G \varphi \equiv$
- $\varphi R \psi \equiv$
- $A \varphi \equiv$

CTL*—Basic Operators

❖ Provided that $AP \neq \emptyset$, it is easy to see that all CTL* operators can be derived from $\vee, \neg, X, U$, and E :

- let $p \in AP$, $true \equiv p \vee \neg p$ (and $false \equiv \neg true$),
- $\varphi \wedge \psi \equiv \neg(\neg\varphi \vee \neg\psi)$
- $F \varphi \equiv$
- $G \varphi \equiv$
- $\varphi R \psi \equiv$
- $A \varphi \equiv$

CTL*—Basic Operators

❖ Provided that $AP \neq \emptyset$, it is easy to see that all CTL* operators can be derived from $\vee, \neg, X, U$, and E :

- let $p \in AP$, $true \equiv p \vee \neg p$ (and $false \equiv \neg true$),
- $\varphi \wedge \psi \equiv \neg(\neg\varphi \vee \neg\psi)$
- $F \varphi \equiv true U \varphi$,
- $G \varphi \equiv$
- $\varphi R \psi \equiv$
- $A \varphi \equiv$

CTL*—Basic Operators

❖ Provided that $AP \neq \emptyset$, it is easy to see that all CTL* operators can be derived from $\vee, \neg, X, U$, and E :

- let $p \in AP$, $true \equiv p \vee \neg p$ (and $false \equiv \neg true$),
- $\varphi \wedge \psi \equiv \neg(\neg\varphi \vee \neg\psi)$
- $F \varphi \equiv true U \varphi$,
- $G \varphi \equiv \neg F \neg\varphi$,
- $\varphi R \psi \equiv$
- $A \varphi \equiv$

CTL*—Basic Operators

❖ Provided that $AP \neq \emptyset$, it is easy to see that all CTL* operators can be derived from $\vee, \neg, X, U$, and E :

- let $p \in AP$, $true \equiv p \vee \neg p$ (and $false \equiv \neg true$),
- $\varphi \wedge \psi \equiv \neg(\neg\varphi \vee \neg\psi)$
- $F \varphi \equiv true U \varphi$,
- $G \varphi \equiv \neg F \neg\varphi$,
- $\varphi R \psi \equiv \neg(\neg\varphi U \neg\psi)$,
- $A \varphi \equiv$

CTL*—Basic Operators

❖ Provided that $AP \neq \emptyset$, it is easy to see that all CTL* operators can be derived from $\vee, \neg, X, U$, and E :

- let $p \in AP$, $true \equiv p \vee \neg p$ (and $false \equiv \neg true$),
- $\varphi \wedge \psi \equiv \neg(\neg\varphi \vee \neg\psi)$
- $F \varphi \equiv true U \varphi$,
- $G \varphi \equiv \neg F \neg\varphi$,
- $\varphi R \psi \equiv \neg(\neg\varphi U \neg\psi)$,
- $A \varphi \equiv \neg E \neg\varphi$.

CTL*—Basic Operators

❖ Provided that $AP \neq \emptyset$, it is easy to see that all CTL* operators can be derived from $\vee, \neg, X, U$, and E :

- let $p \in AP$, $true \equiv p \vee \neg p$ (and $false \equiv \neg true$),
- $\varphi \wedge \psi \equiv \neg(\neg\varphi \vee \neg\psi)$
- $F \varphi \equiv true U \varphi$,
- $G \varphi \equiv \neg F \neg\varphi$,
- $\varphi R \psi \equiv \neg(\neg\varphi U \neg\psi)$,
- $A \varphi \equiv \neg E \neg\varphi$.

❖ Some further connectives may be introduced too, e.g.:

- $\varphi \Rightarrow \psi \equiv \neg\varphi \vee \psi$,
- $\varphi \Leftrightarrow \psi \equiv (\varphi \Rightarrow \psi) \wedge (\psi \Rightarrow \varphi)$,
- ...

The CTL Logic

CTL—The Syntax

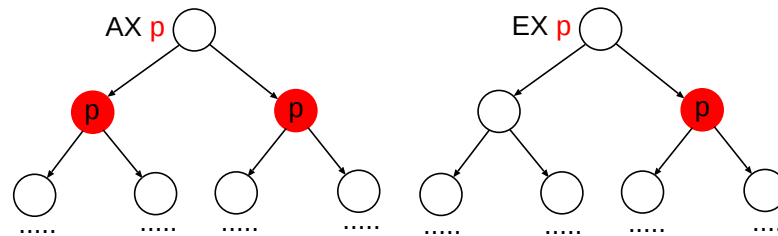
❖ CTL is a sublogic of CTL*—the path formulae are restricted to $X \varphi$, $F \varphi$, $G \varphi$, $\varphi U \psi$, and $\varphi R \psi$ for φ, ψ being state formulae.

CTL—The Syntax

❖ CTL is a sublogic of CTL*—the path formulae are restricted to $X \varphi$, $F \varphi$, $G \varphi$, $\varphi U \psi$, and $\varphi R \psi$ for φ, ψ being state formulae.

❖ In effect, there are allowed these 10 modal CTL operators:

- AX and EX ,



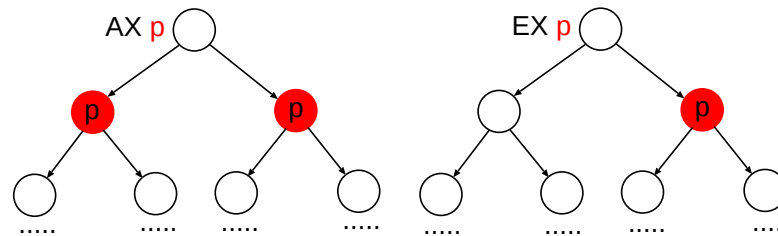
Continued at the next slide...

CTL—The Syntax

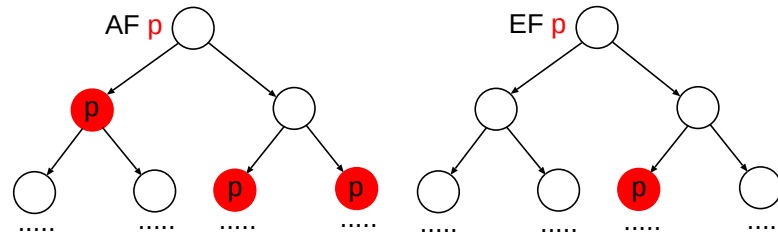
❖ CTL is a sublogic of CTL*—the path formulae are restricted to $X \varphi$, $F \varphi$, $G \varphi$, $\varphi U \psi$, and $\varphi R \psi$ for φ, ψ being state formulae.

❖ In effect, there are allowed these 10 modal CTL operators:

- AX and EX ,



- AF and EF ,



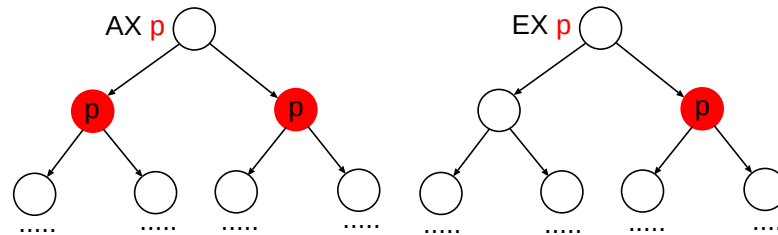
Continued at the next slide...

CTL—The Syntax

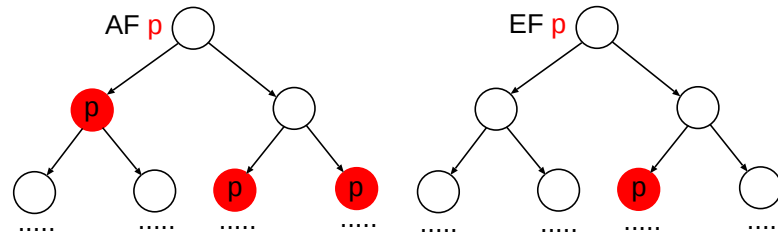
❖ CTL is a sublogic of CTL*—the path formulae are restricted to $X \varphi$, $F \varphi$, $G \varphi$, $\varphi U \psi$, and $\varphi R \psi$ for φ, ψ being state formulae.

❖ In effect, there are allowed these 10 modal CTL operators:

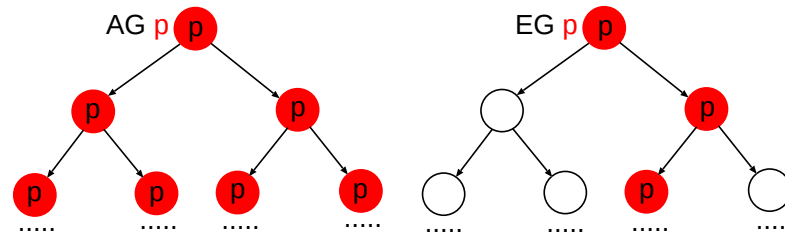
- AX and EX ,



- AF and EF ,



- AG and EG ,

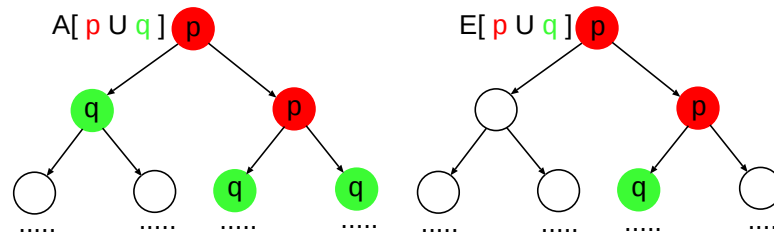


Continued at the next slide...

CTL—The Syntax

Continued from the previous slide...

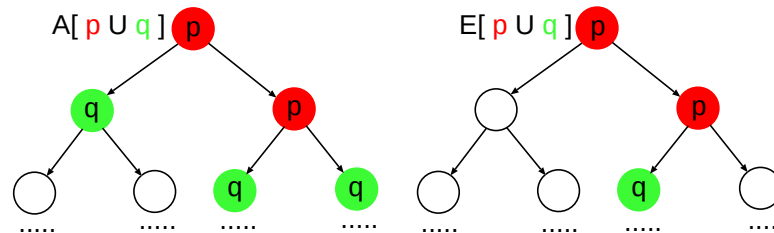
- AU and EU ,



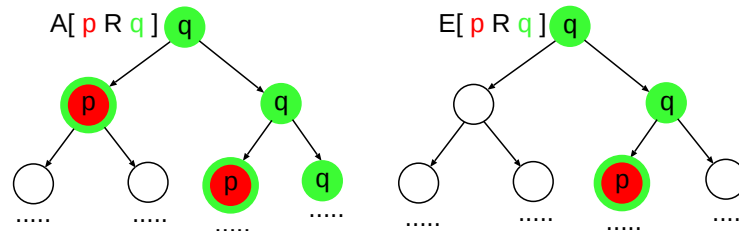
CTL—The Syntax

Continued from the previous slide...

- AU and EU ,



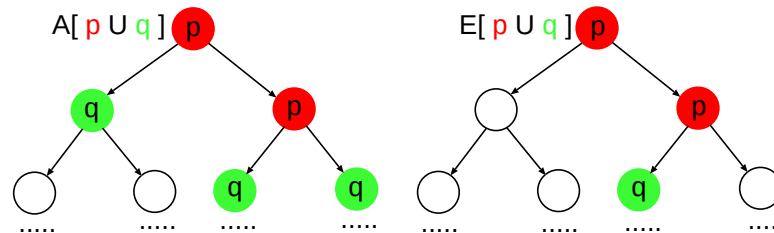
- AR and ER .



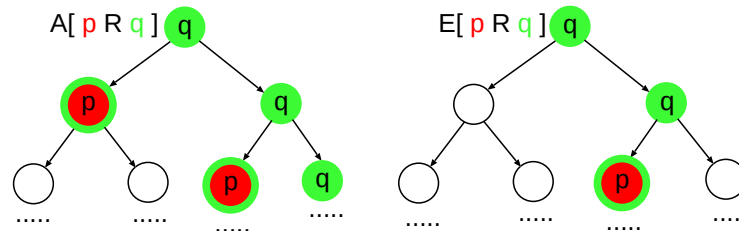
CTL—The Syntax

Continued from the previous slide...

- AU and EU ,



- AR and ER .



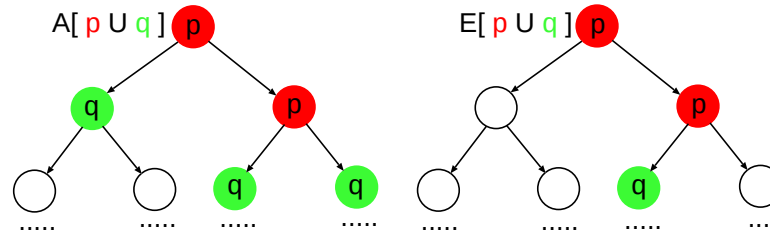
❖ There are 3 basic CTL modal operators— EX , EG , and EU :

- $AX \varphi \equiv$
- $EF \varphi \equiv$
- $AG \varphi \equiv$
- $AF \varphi \equiv$
- $A[\varphi U \psi] \equiv$
- $A[\varphi R \psi] \equiv$
- $E[\varphi R \psi] \equiv$

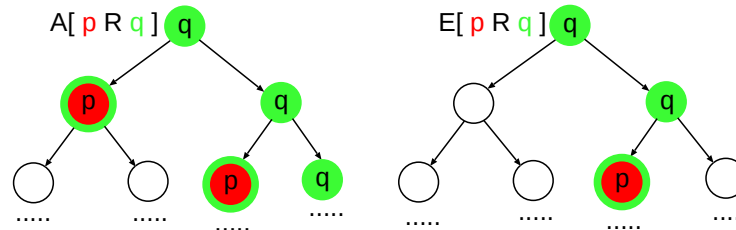
CTL—The Syntax

Continued from the previous slide...

- AU and EU ,



- AR and ER .



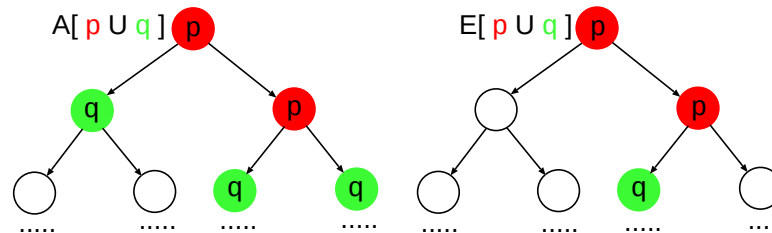
❖ There are 3 basic CTL modal operators— EX , EG , and EU :

- $AX \varphi \equiv \neg EX \neg \varphi$,
- $EF \varphi \equiv$
- $AG \varphi \equiv$
- $AF \varphi \equiv$
- $A[\varphi U \psi] \equiv$
- $A[\varphi R \psi] \equiv$
- $E[\varphi R \psi] \equiv$

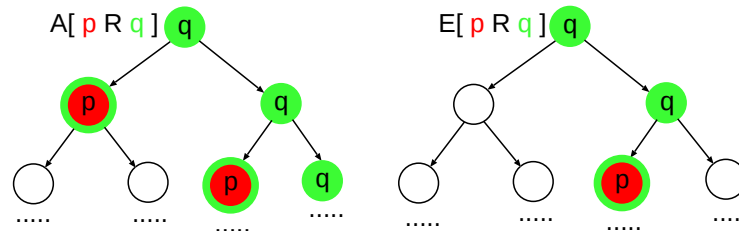
CTL—The Syntax

Continued from the previous slide...

- AU and EU ,



- AR and ER .



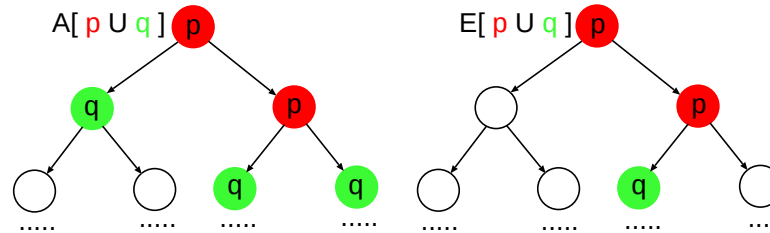
❖ There are 3 basic CTL modal operators— EX , EG , and EU :

- $AX \varphi \equiv \neg EX \neg \varphi$,
- $EF \varphi \equiv E[true U \varphi]$,
- $AG \varphi \equiv$
- $AF \varphi \equiv$
- $A[\varphi U \psi] \equiv$
- $A[\varphi R \psi] \equiv$
- $E[\varphi R \psi] \equiv$

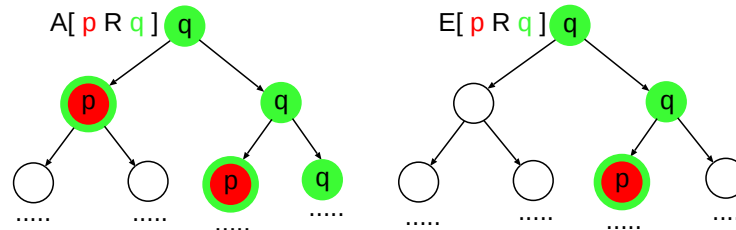
CTL—The Syntax

Continued from the previous slide...

- AU and EU ,



- AR and ER .



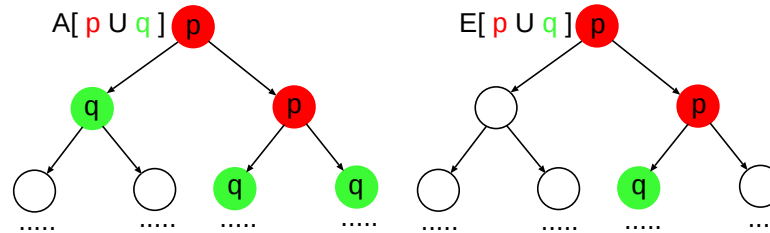
❖ There are 3 basic CTL modal operators— EX , EG , and EU :

- $AX \varphi \equiv \neg EX \neg \varphi$,
- $EF \varphi \equiv E[true U \varphi]$,
- $AG \varphi \equiv \neg EF \neg \varphi$,
- $AF \varphi \equiv$
- $A[\varphi U \psi] \equiv$
- $A[\varphi R \psi] \equiv$
- $E[\varphi R \psi] \equiv$

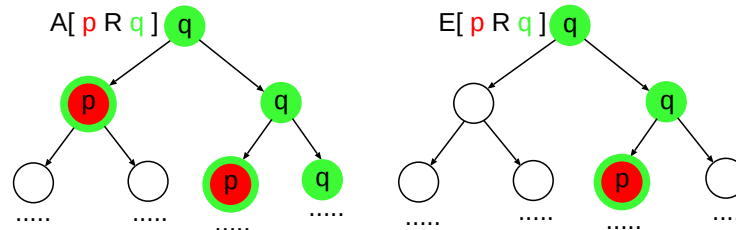
CTL—The Syntax

Continued from the previous slide...

- AU and EU ,



- AR and ER .



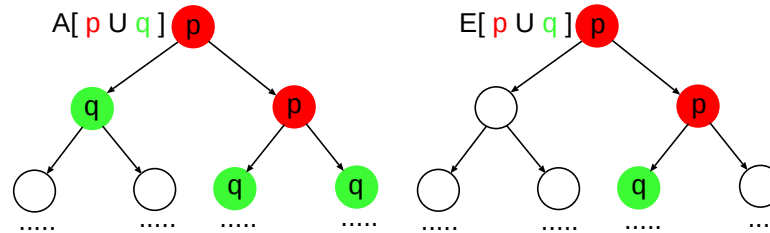
❖ There are 3 basic CTL modal operators— EX , EG , and EU :

- $AX \varphi \equiv \neg EX \neg \varphi$,
- $EF \varphi \equiv E[true U \varphi]$,
- $AG \varphi \equiv \neg EF \neg \varphi$,
- $AF \varphi \equiv \neg EG \neg \varphi$,
- $A[\varphi U \psi] \equiv$
- $A[\varphi R \psi] \equiv$
- $E[\varphi R \psi] \equiv$

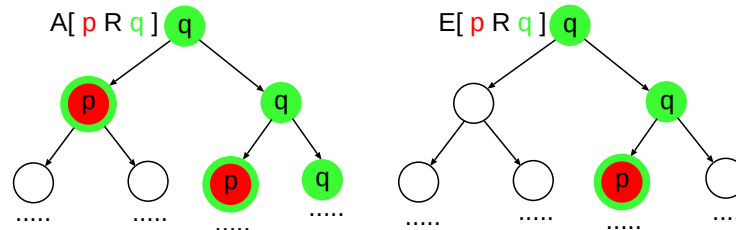
CTL—The Syntax

Continued from the previous slide...

- AU and EU ,



- AR and ER .



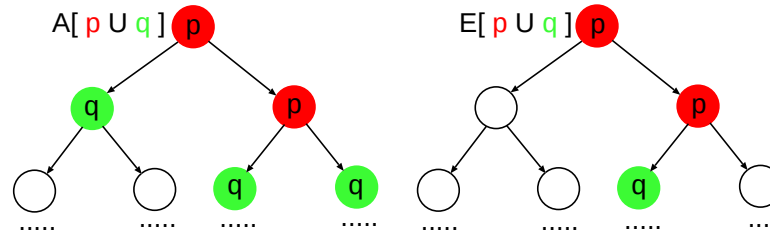
❖ There are 3 basic CTL modal operators— EX , EG , and EU :

- $AX \varphi \equiv \neg EX \neg \varphi$,
- $EF \varphi \equiv E[true U \varphi]$,
- $AG \varphi \equiv \neg EF \neg \varphi$,
- $AF \varphi \equiv \neg EG \neg \varphi$,
- $A[\varphi U \psi] \equiv \neg E[\neg \psi U (\neg \varphi \wedge \neg \psi)] \wedge AF \psi$,
- $A[\varphi R \psi] \equiv$
- $E[\varphi R \psi] \equiv$

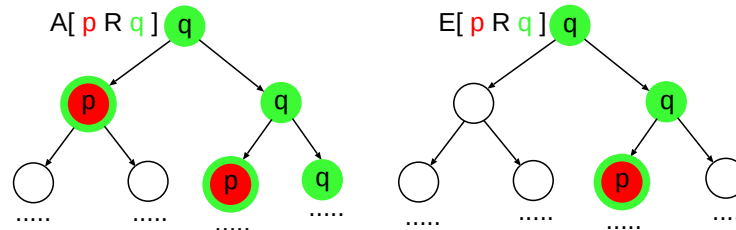
CTL—The Syntax

Continued from the previous slide...

- AU and EU ,



- AR and ER .



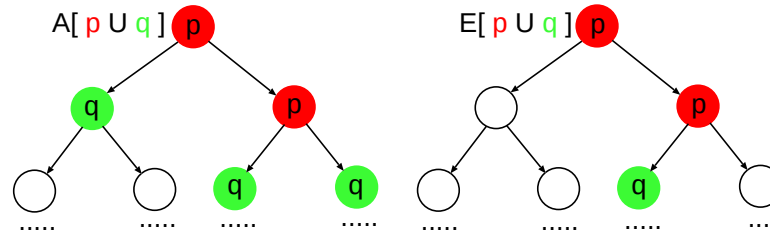
❖ There are 3 basic CTL modal operators— EX , EG , and EU :

- $AX \ \varphi \equiv \neg EX \ \neg \varphi$,
- $EF \ \varphi \equiv E[true \ U \ \varphi]$,
- $AG \ \varphi \equiv \neg EF \ \neg \varphi$,
- $AF \ \varphi \equiv \neg EG \ \neg \varphi$,
- $A[\varphi \ U \ \psi] \equiv \neg E[\neg \psi \ U \ (\neg \varphi \wedge \neg \psi)] \wedge AF \psi$,
- $A[\varphi \ R \ \psi] \equiv \neg E[\neg \varphi \ U \ \neg \psi]$,
- $E[\varphi \ R \ \psi] \equiv$

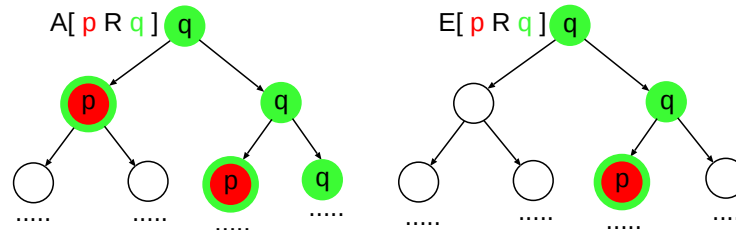
CTL—The Syntax

Continued from the previous slide...

- AU and EU ,



- AR and ER .



❖ There are 3 basic CTL modal operators— EX , EG , and EU :

- $AX \varphi \equiv \neg EX \neg \varphi$,
- $EF \varphi \equiv E[true U \varphi]$,
- $AG \varphi \equiv \neg EF \neg \varphi$,
- $AF \varphi \equiv \neg EG \neg \varphi$,
- $A[\varphi U \psi] \equiv \neg E[\neg \psi U (\neg \varphi \wedge \neg \psi)] \wedge AF \psi$,
- $A[\varphi R \psi] \equiv \neg E[\neg \varphi U \neg \psi]$,
- $E[\varphi R \psi] \equiv \neg A[\neg \varphi U \neg \psi]$.

CTL—Some Examples

❖ Some examples of CTL formulae:

- Mutual exclusion of two processes using propositions $cs(P1)$ (process $P1$ is in the critical section) and $cs(P2)$.

CTL—Some Examples

❖ Some examples of CTL formulae:

- Mutual exclusion of two processes using propositions $cs(P1)$ (process $P1$ is in the critical section) and $cs(P2)$.

$$\neg EF (cs(P1) \wedge cs(P2)) \equiv AG (\neg cs(P1) \vee \neg cs(P2))$$

CTL—Some Examples

❖ Some examples of CTL formulae:

- Mutual exclusion of two processes using propositions $cs(P1)$ (process $P1$ is in the critical section) and $cs(P2)$.
$$\neg EF (cs(P1) \wedge cs(P2)) \equiv AG (\neg cs(P1) \vee \neg cs(P2))$$
- It is possible to get to a state where $Start$ holds, but $Ready$ does not.

CTL—Some Examples

❖ Some examples of CTL formulae:

- Mutual exclusion of two processes using propositions $cs(P1)$ (process $P1$ is in the critical section) and $cs(P2)$.

$$\neg EF (cs(P1) \wedge cs(P2)) \equiv AG (\neg cs(P1) \vee \neg cs(P2))$$

- It is possible to get to a state where $Start$ holds, but $Ready$ does not.

$$EF (Start \wedge \neg Ready)$$

CTL—Some Examples

❖ Some examples of CTL formulae:

- Mutual exclusion of two processes using propositions $cs(P1)$ (process $P1$ is in the critical section) and $cs(P2)$.
$$\neg EF (cs(P1) \wedge cs(P2)) \equiv AG (\neg cs(P1) \vee \neg cs(P2))$$
- It is possible to get to a state where $Start$ holds, but $Ready$ does not.
$$EF (Start \wedge \neg Ready)$$
- If a request occurs (i.e. Req holds), then it will eventually be acknowledged (i.e. Ack will hold).

CTL—Some Examples

❖ Some examples of CTL formulae:

- Mutual exclusion of two processes using propositions $cs(P1)$ (process $P1$ is in the critical section) and $cs(P2)$.
$$\neg EF (cs(P1) \wedge cs(P2)) \equiv AG (\neg cs(P1) \vee \neg cs(P2))$$
- It is possible to get to a state where $Start$ holds, but $Ready$ does not.
$$EF (Start \wedge \neg Ready)$$
- If a request occurs (i.e. Req holds), then it will eventually be acknowledged (i.e. Ack will hold).
$$AG (Req \Rightarrow AF Ack)$$

CTL—Some Examples

❖ Some examples of CTL formulae:

- Mutual exclusion of two processes using propositions $cs(P1)$ (process $P1$ is in the critical section) and $cs(P2)$.
$$\neg EF (cs(P1) \wedge cs(P2)) \equiv AG (\neg cs(P1) \vee \neg cs(P2))$$
- It is possible to get to a state where $Start$ holds, but $Ready$ does not.
$$EF (Start \wedge \neg Ready)$$
- If a request occurs (i.e. Req holds), then it will eventually be acknowledged (i.e. Ack will hold).
$$AG (Req \Rightarrow AF Ack)$$
- In any run of the system, $DeviceEnabled$ is true infinitely often.

CTL—Some Examples

❖ Some examples of CTL formulae:

- Mutual exclusion of two processes using propositions $cs(P1)$ (process $P1$ is in the critical section) and $cs(P2)$.
 $\neg EF (cs(P1) \wedge cs(P2)) \equiv AG (\neg cs(P1) \vee \neg cs(P2))$
- It is possible to get to a state where $Start$ holds, but $Ready$ does not.
 $EF (Start \wedge \neg Ready)$
- If a request occurs (i.e. Req holds), then it will eventually be acknowledged (i.e. Ack will hold).
 $AG (Req \Rightarrow AF Ack)$
- In any run of the system, $DeviceEnabled$ is true infinitely often.
 $AG AF DeviceEnabled$

CTL—Some Examples

❖ Some examples of CTL formulae:

- Mutual exclusion of two processes using propositions $cs(P1)$ (process $P1$ is in the critical section) and $cs(P2)$.
 $\neg EF (cs(P1) \wedge cs(P2)) \equiv AG (\neg cs(P1) \vee \neg cs(P2))$
- It is possible to get to a state where $Start$ holds, but $Ready$ does not.
 $EF (Start \wedge \neg Ready)$
- If a request occurs (i.e. Req holds), then it will eventually be acknowledged (i.e. Ack will hold).
 $AG (Req \Rightarrow AF Ack)$
- In any run of the system, $DeviceEnabled$ is true infinitely often.
 $AG AF DeviceEnabled$
- From any state, the system can be restarted (i.e. get to a $Restart$ state).

CTL—Some Examples

❖ Some examples of CTL formulae:

- Mutual exclusion of two processes using propositions $cs(P1)$ (process $P1$ is in the critical section) and $cs(P2)$.
 $\neg EF (cs(P1) \wedge cs(P2)) \equiv AG (\neg cs(P1) \vee \neg cs(P2))$
- It is possible to get to a state where $Start$ holds, but $Ready$ does not.
 $EF (Start \wedge \neg Ready)$
- If a request occurs (i.e. Req holds), then it will eventually be acknowledged (i.e. Ack will hold).
 $AG (Req \Rightarrow AF Ack)$
- In any run of the system, $DeviceEnabled$ is true infinitely often.
 $AG AF DeviceEnabled$
- From any state, the system can be restarted (i.e. get to a $Restart$ state).
 $AG EF Restart$

CTL—Some Examples

❖ Some examples of CTL formulae:

- Mutual exclusion of two processes using propositions $cs(P1)$ (process $P1$ is in the critical section) and $cs(P2)$.
 $\neg EF (cs(P1) \wedge cs(P2)) \equiv AG (\neg cs(P1) \vee \neg cs(P2))$
- It is possible to get to a state where $Start$ holds, but $Ready$ does not.
 $EF (Start \wedge \neg Ready)$
- If a request occurs (i.e. Req holds), then it will eventually be acknowledged (i.e. Ack will hold).
 $AG (Req \Rightarrow AF Ack)$
- In any run of the system, $DeviceEnabled$ is true infinitely often.
 $AG AF DeviceEnabled$
- From any state, the system can be restarted (i.e. get to a $Restart$ state).
 $AG EF Restart$
- The $Reset$ signal is initially set, and from the next state on, it is never set again.

CTL—Some Examples

❖ Some examples of CTL formulae:

- Mutual exclusion of two processes using propositions $cs(P1)$ (process $P1$ is in the critical section) and $cs(P2)$.
 $\neg EF (cs(P1) \wedge cs(P2)) \equiv AG (\neg cs(P1) \vee \neg cs(P2))$
- It is possible to get to a state where $Start$ holds, but $Ready$ does not.
 $EF (Start \wedge \neg Ready)$
- If a request occurs (i.e. Req holds), then it will eventually be acknowledged (i.e. Ack will hold).
 $AG (Req \Rightarrow AF Ack)$
- In any run of the system, $DeviceEnabled$ is true infinitely often.
 $AG AF DeviceEnabled$
- From any state, the system can be restarted (i.e. get to a $Restart$ state).
 $AG EF Restart$
- The $Reset$ signal is initially set, and from the next state on, it is never set again.
 $Reset \wedge AX AG \neg Reset$

CTL—Some Examples

❖ Some examples of CTL formulae:

- Mutual exclusion of two processes using propositions $cs(P1)$ (process $P1$ is in the critical section) and $cs(P2)$.
 $\neg EF (cs(P1) \wedge cs(P2)) \equiv AG (\neg cs(P1) \vee \neg cs(P2))$
- It is possible to get to a state where $Start$ holds, but $Ready$ does not.
 $EF (Start \wedge \neg Ready)$
- If a request occurs (i.e. Req holds), then it will eventually be acknowledged (i.e. Ack will hold).
 $AG (Req \Rightarrow AF Ack)$
- In any run of the system, $DeviceEnabled$ is true infinitely often.
 $AG AF DeviceEnabled$
- From any state, the system can be restarted (i.e. get to a $Restart$ state).
 $AG EF Restart$
- The $Reset$ signal is initially set, and from the next state on, it is never set again.
 $Reset \wedge AX AG \neg Reset$
- The $Reset$ signal is initially set, but once it is unset, it is never set again.

CTL—Some Examples

❖ Some examples of CTL formulae:

- Mutual exclusion of two processes using propositions $cs(P1)$ (process $P1$ is in the critical section) and $cs(P2)$.
 $\neg EF (cs(P1) \wedge cs(P2)) \equiv AG (\neg cs(P1) \vee \neg cs(P2))$
- It is possible to get to a state where $Start$ holds, but $Ready$ does not.
 $EF (Start \wedge \neg Ready)$
- If a request occurs (i.e. Req holds), then it will eventually be acknowledged (i.e. Ack will hold).
 $AG (Req \Rightarrow AF Ack)$
- In any run of the system, $DeviceEnabled$ is true infinitely often.
 $AG AF DeviceEnabled$
- From any state, the system can be restarted (i.e. get to a $Restart$ state).
 $AG EF Restart$
- The $Reset$ signal is initially set, and from the next state on, it is never set again.
 $Reset \wedge AX AG \neg Reset$
- The $Reset$ signal is initially set, but once it is unset, it is never set again.
 $Reset \wedge AG (\neg Reset \Rightarrow AG \neg Reset)$

CTL—Some Examples

❖ Some examples of CTL formulae:

- Mutual exclusion of two processes using propositions $cs(P1)$ (process $P1$ is in the critical section) and $cs(P2)$.
 $\neg EF (cs(P1) \wedge cs(P2)) \equiv AG (\neg cs(P1) \vee \neg cs(P2))$
- It is possible to get to a state where $Start$ holds, but $Ready$ does not.
 $EF (Start \wedge \neg Ready)$
- If a request occurs (i.e. Req holds), then it will eventually be acknowledged (i.e. Ack will hold).
 $AG (Req \Rightarrow AF Ack)$
- In any run of the system, $DeviceEnabled$ is true infinitely often.
 $AG AF DeviceEnabled$
- From any state, the system can be restarted (i.e. get to a $Restart$ state).
 $AG EF Restart$
- The $Reset$ signal is initially set, and from the next state on, it is never set again.
 $Reset \wedge AX AG \neg Reset$
- The $Reset$ signal is initially set, but once it is unset, it is never set again.
 $Reset \wedge AG (\neg Reset \Rightarrow AG \neg Reset)$
- The $AccConn$ signal can be set only after the $StartAcc$ signal arrives.

CTL—Some Examples

❖ Some examples of CTL formulae:

- Mutual exclusion of two processes using propositions $cs(P1)$ (process $P1$ is in the critical section) and $cs(P2)$.
 $\neg EF (cs(P1) \wedge cs(P2)) \equiv AG (\neg cs(P1) \vee \neg cs(P2))$
- It is possible to get to a state where $Start$ holds, but $Ready$ does not.
 $EF (Start \wedge \neg Ready)$
- If a request occurs (i.e. Req holds), then it will eventually be acknowledged (i.e. Ack will hold).
 $AG (Req \Rightarrow AF Ack)$
- In any run of the system, $DeviceEnabled$ is true infinitely often.
 $AG AF DeviceEnabled$
- From any state, the system can be restarted (i.e. get to a $Restart$ state).
 $AG EF Restart$
- The $Reset$ signal is initially set, and from the next state on, it is never set again.
 $Reset \wedge AX AG \neg Reset$
- The $Reset$ signal is initially set, but once it is unset, it is never set again.
 $Reset \wedge AG (\neg Reset \Rightarrow AG \neg Reset)$
- The $AccConn$ signal can be set only after the $StartAcc$ signal arrives.
 $A[StartAcc R (\neg AccConn)]$

CTL Model Checking

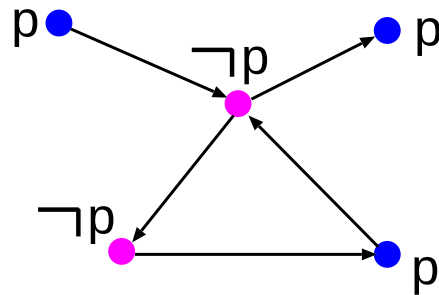
The Basic Idea

- ❖ The **CTL model checking question** to be answered: Given a Kripke structure $M = (S, S_0, R, L)$ over a set of atomic propositions AP and a CTL formula φ over AP , does $M \models \varphi$ hold?
- ❖ A **very basic approach** to answer the CTL model checking question by the so-called **explicit-state model checking**:
 - For every **subformula** ψ of φ , **label by** ψ all those states s of M in which ψ holds (i.e., $M, s \models \psi$).
 - Perform the labelling **from the inner-most subformulae** (i.e. the most nested ones) going **to the outer ones** exploiting the already computed labels (with atomic propositions corresponding to the original labels of M).
 - Check whether each state in S_0 gets labelled by φ .
- ❖ It is enough to consider the **basic operators** of CTL, i.e. the below syntax for $p \in AP$:
$$\varphi ::= p \mid \neg\varphi \mid \varphi \vee \varphi \mid EX\varphi \mid E[\varphi U \varphi] \mid EG\varphi.$$

$Label(\neg\varphi), Label(\varphi_1 \vee \varphi_2)$

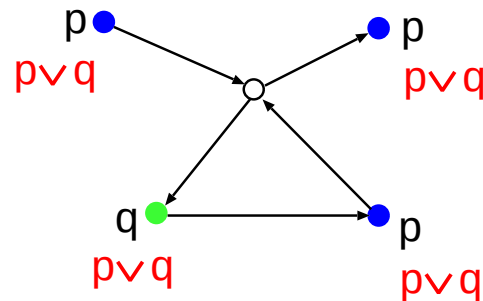
$Label(\neg\varphi)$

for all $s \in S$ such that $\varphi \notin Label(s)$ do
 $Label(s) := Label(s) \cup \{\neg\varphi\}$



$Label(\varphi_1 \vee \varphi_2)$

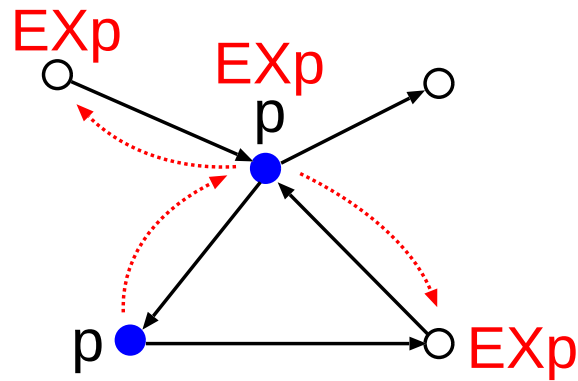
for all $s \in S$ such that $\varphi_1 \in Label(s)$ or $\varphi_2 \in Label(s)$ do
 $Label(s) := Label(s) \cup \{\varphi_1 \vee \varphi_2\}$



Label($EX\varphi$)

Label($EX\varphi$)

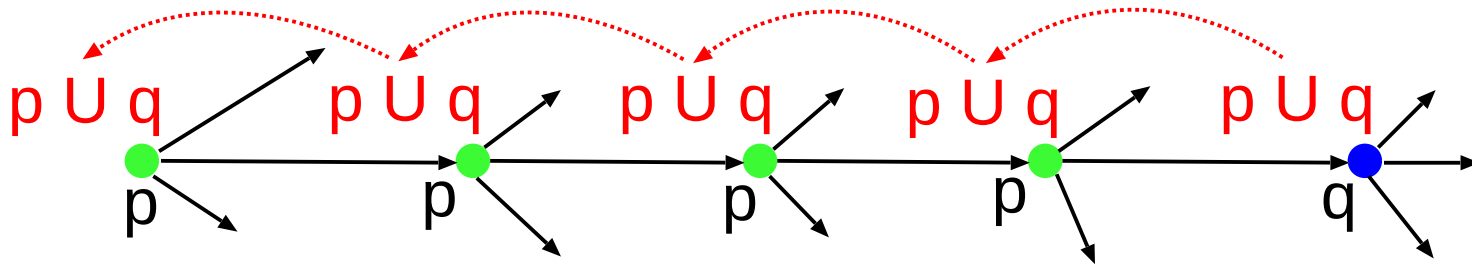
for all $s_2 \in S$ such that $\varphi \in Label(s_2)$ do
 for all $s_1 \in S$ such that $R(s_1, s_2)$ do
 $Label(s_1) := Label(s_1) \cup \{EX\varphi\}$



Label($E[\varphi_1 U \varphi_2]$)

❖ The idea:

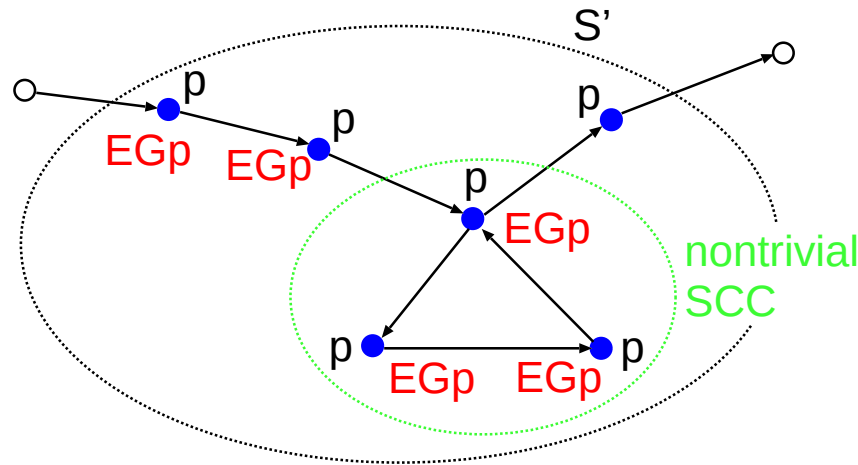
- Label first states already labelled by φ_2 .
- Look at **predecessors** of states labelled with $\varphi_1 U \varphi_2$, and if they are labelled with φ_1 , label them with $\varphi_1 U \varphi_2$ as well.



Label($EG\varphi$)

❖ Based on the following observation: Let $M = (S, S_0, R, L)$ be a Kripke structure, $S' = \{s \in S \mid M, s \models \varphi\}$, and $R' = R \cap (S' \times S')$. For any $s \in S$, $M, s \models EG\varphi$ iff

1. $s \in S'$ and
2. there exists a **path** in the oriented graph $G' = (S', R')$ that leads from s to some node t in a **nontrivial SCC** C of G' .



❖ An SCC C is **nontrivial** iff either it has more than one node or it contains one node with a self-loop.

❖ SCCs of a finite oriented graph (V, E) can be computed using the **Tarjan's algorithm** in time $O(|E| + |V|)$.

The LTL Logic

LTL—The Syntax

❖ LTL is another sublogic of CTL* that allows only formulae of the form $A \varphi$ in which the only state subformulae are atomic propositions.

❖ This is, LTL formulae φ are built according to the grammar:

- $\varphi ::= A \psi$ (the use of A is often omitted),
- $\psi ::= p \mid \neg\psi \mid \psi \vee \psi \mid \psi \wedge \psi \mid X \psi \mid F \psi \mid G \psi \mid \psi U \psi \mid \psi R \psi$

where $p \in AP$.

❖ Note that LTL speaks about particular paths in a given Kripke structure only—it ignores its branching structure.

❖ Sometimes, existential LTL allowing formulae of the form $E \varphi$ is used too.

LTL—The Syntax

❖ LTL is another sublogic of CTL* that allows only formulae of the form $A \varphi$ in which the only state subformulae are atomic propositions.

❖ This is, LTL formulae φ are built according to the grammar:

- $\varphi ::= A \psi$ (the use of A is often omitted),
- $\psi ::= p \mid \neg\psi \mid \psi \vee \psi \mid \psi \wedge \psi \mid X \psi \mid F \psi \mid G \psi \mid \psi U \psi \mid \psi R \psi$

where $p \in AP$.

❖ Note that LTL speaks about particular paths in a given Kripke structure only—it ignores its branching structure.

❖ Sometimes, existential LTL allowing formulae of the form $E \varphi$ is used too.

❖ Note also that while CTL cannot express fairness assumptions (in CTL model checking, they are handled by a special extension of the model checking algorithm), LTL can express fairness assumptions by formulae of the following form:

- **weak fairness:** $(F G \text{ Enabled}) \Rightarrow (G F \text{ Fired})$, i.e. $\Diamond \Box \text{ Enabled} \Rightarrow \Box \Diamond \text{ Fired}$,
- **strong fairness:** $(G F \text{ Enabled}) \Rightarrow (G F \text{ Fired})$, i.e. $\Box \Diamond \text{ Enabled} \Rightarrow \Box \Diamond \text{ Fired}$.

LTL, CTL, and CTL*

❖ LTL and CTL have an incomparable power:

- CTL cannot express, e.g., the LTL formula $A (FG\ p)$,
- LTL cannot express, e.g., the CTL formula $AG (EF\ p)$.

LTL, CTL, and CTL*

❖ LTL and CTL have an incomparable power:

- CTL cannot express, e.g., the LTL formula $A (FG p)$,
- LTL cannot express, e.g., the CTL formula $AG (EF p)$.

❖ CTL* is strictly more powerful than both LTL and CTL:

- the disjunction of the above formulae, i.e. $(A (FG p)) \vee (AG (EF p))$, is not expressible in CTL nor LTL.

LTL, CTL, and CTL*

❖ LTL and CTL have an incomparable power:

- CTL cannot express, e.g., the LTL formula $A (FG p)$,
- LTL cannot express, e.g., the CTL formula $AG (EF p)$.

❖ CTL* is strictly more powerful than both LTL and CTL:

- the disjunction of the above formulae, i.e. $(A (FG p)) \vee (AG (EF p))$, is not expressible in CTL nor LTL.

❖ To complete the picture, here are the complexities of the appropriate model checking algorithms (we will discuss LTL model checking later on):

- CTL: linear in $|M|$ and linear in $|\varphi|$.
- LTL and CTL*: linear in $|M|$ and PSPACE-complete in $|\varphi|$

where $|M| = |S| + |R|$ and $|\varphi|$ is the number of subformulae of φ .

❖ Finally, as an example of a logic more general than CTL*, we can mention modal μ -calculus based on least/greatest fixpoint operators on sets of states (basically allowing one to define new, specialised modalities).